USER MANUAL



Lithium battery charger

Welcome to use our charger products. Please read the "User Manual" thoroughly for customers to use the charger quickly and safely.

★ Attention

The manual provides charger parameters, installation, fault diagnosis, and daily maintenance precautions. To ensure correct installation and operation of this charger, please read the manual in detail before installation and keep it properly..

★ Special notice

★ The content of this manual is subject to change without prior notice.

★ If any errors or omissions are found in the manual, please contact the manufacturer.

★ We are not responsible for any direct or indirect losses caused by improper use not in accordance with this manual.

★ Unauthorized conversion or reproduction of all or part of this manual is prohibited.

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Chapter 1 Overview

1.1 Instructions for Use

The safety precautions, product features, technical performance, operation steps, installation and debugging, fault handling, equipment maintenance, and other contents of the lithium battery charger are expounded and explained. Please read the manual carefully after opening the box, and keep it properly for later reference.

The Company reserves the right to modify the specifications and has the right not to give further notice.

1.2 Safety warning



1.3 Terms of Use

★ When using and connecting the lithium battery charger, the relevant regulations for electric forklift trucks charging should be strictly observed;

★ The lithium battery charger has been developed, produced,

inspected and filed according to the relevant safety standards. If all prescribed instructions and safety technical instructions are followed, no property damage or harm to personnelonal safety is normally caused;

★The instructions contained in this manual must be strictly followed, otherwise there may be a safety hazard or cause the failure of the safety device. Although the relevant safety instructions are described in this manual, care must be taken to comply with the safety regulations and accident prevention regulations in the area where the product is used.

1.4 For the crowd

this manual is intended for the following groups of people:

★ Charging equipment training personnel;
★ Charging operator;
★ Technical service personnel .

chapter2 safety precautions

2.1safety statement

Failure to comply with the safety precautions in this section of the use of the equipment may lead to life danger, injury and equipment damage, Hangzhou Hangcha Yunrui Technology Co., Ltd. shall not be liable for any claims arising therefrom.

2.2safety tips

To ensure personal and equipment safety, the personnel operating the equipment must comply with the following principles and regulations:

★ Before the equipment is powered on, please confirm that the equipment is well grounded;

★ Before the equipment is energized, confirm that the input voltage, frequency, circuit breaker or fuse and other conditions of the device meet the specifications;

★All tools should be insulated to prevent exposed metal parts from touching the metal frame, resulting in short circuit ;

★ Ensure the stable operation of the equipment, the equipment shall operate in a clean and well-ventilated environment as far as possible, and the operating environment shall not contain volatile gases or flammable gases.

2.3safety regulations

number	safety signs		meaning
1		Earthing identification	It represents the installation position of the grounding wire.

1.danger

★ Electrical hazard, fire hazard! Do not use damaged, worn, or dirty charging guns;

★Do not place inflammable, explosive or combustible materials, chemicals, combustible steam and other dangerous goods close to the charging equipment;

★The power supply must be cut off during the installation and replacement of the equipment to prevent electric shock;

★Do not use any extension cable to connect the charging device to the electric forklift trucks

the charging device is only connected to the electric forklift trucks, do not connect to other loads (power tools), etc.

★Do not cause mechanical damage to the charging cable (bending, squeezing or rolling), and The surface of the cable should not come into contact with heat sources, dirt, or water sources

★All cables of the equipment must be firmly wired, well insulated and properly sized. Loose connection, damage or insufficient size of the cable will cause circuit failure, serious may cause irreversible damage to the equipment, and even cause casualties or fire;

★Falling equipment, even if the appearance is not damaged, also can not be directly installed, because part of the internal damage may cause system disorder or circuit short circuit and other faults, must be tested by professional personnel, to determine whether it can be installed;

★Never allowed to flush the equipment with water, the equipment

into the water will cause a short circuit.

2. warn

★Any fault that may affect its safety performance must be immediately eliminated by a professional before it can be restarted;

★ When pulling out the charging gun, please hold the charging gun handle, do not pull the cable;

★Do not let heavy objects press down on the cable, or people step on the charging cable;

★Keep the charging gun clean and dry. If there is dirt, please wipe it with a clean dry cloth. Do not touch the charging gun core directly with your hand.

★The products are installed by the company's professional and technical personnel (or an authorized professional organization), and the unauthorized installation or modification may lead to safety accidents or equipment damage. The Company shall not be liable for any loss caused thereby.

3. take care

★ If there is any abnormal situation during use, please immediately press the emergency stop button to cut off all input and output power supply;

★In case of rain and thunder weather, please use it carefully;

★ The normal operation of the equipment depends on the reasonable setting of some parameters, which are generally set after on-site installation and debugging. If not necessary, it is not recommended for users to reset the parameters by themselves;

★Charging can only be carried out after the vehicle is turned off, and the vehicle can only be started after charging is completed. During the charging process, electric forklift trucks are prohibited from driving;

★ Regularly check whether the charging gun and charging cable are damaged, and check whether the equipment shell is damaged;

★When using charging devices to charge electric forklift trucks,

Carefully read the relevant prompts and user manual of the vehicle;

★The warning and operation signs inside the equipment body and cabinet play an important role in the safe operation of the equipment. It is strictly prohibited to tear or damage them. If there is any damage or blurriness, please replace them in a timely manner.

Chapter 3 Product Introduction

3.1 Referenced Standard

★ Q/HCC J03045-2021 Off-board conductive lithium battery charger for industrial vehicles ;

★ T/CCMA 0110-2020 Off-board conductive charger for industrial vehicles ;

★ NB/T 33001-2018 Technical Conditions for Off-Board Conductive Chargers for Electric Forklift Trucks;

★ NB/T 33008.1-2018 Specification for Inspection and Testing of Electric Forklift Trucks Charging Equipment Part 1: Off-board Charging Machines;

★ GB/T 27930-2015 Communication Protocol between Off-Board Conductive Chargers and Battery Management Systems for Electric Forklift Trucks;

3.2 Conditions of Use

★ The altitude shall not exceed 2000 m;

★ Operating environment temperature of the equipment: -20 ℃ ~ + 50℃;

★ The maximum relative humidity of the surrounding air shall not exceed 95% (when the temperature of the ambient air is 25 ± 5℃);

★ The installation foundation should be stable, and there should be no violent vibration or impact, and the vertical inclination should not exceed 5%;

★ The operation site has no conductive or explosive dust, no gas or vapor that corrodes metal or destroys insulation;

★ Outdoor operation requires the installation of protective sheds around and above the equipment to prevent rain.

3.3 Product overview

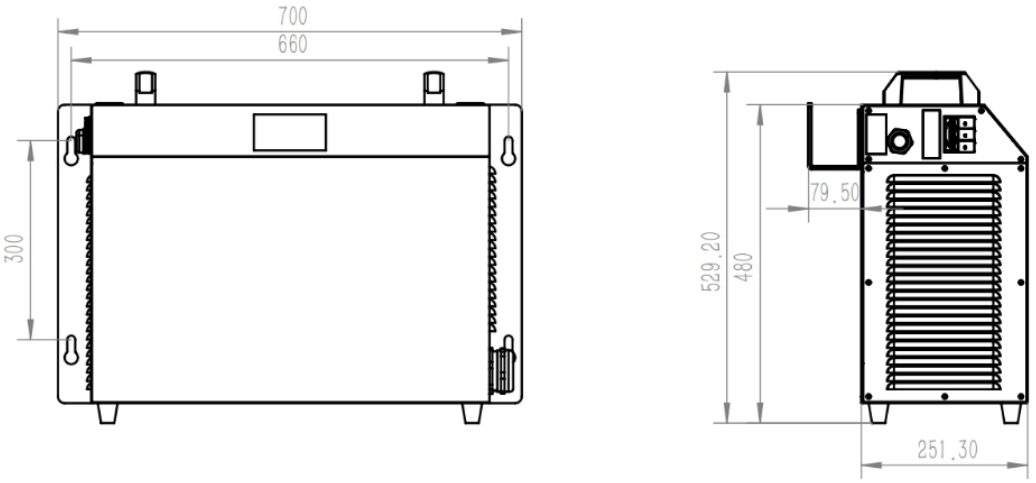
This product has a high input power factor and low input current harmonic. high efficiency, EMC performance, simple technology, high standardization, high reliability, beautiful appearance, simple use, operation, maintenance, far better than the same application products in the industry, can be widely used in transportation, industry, logistics, and other industries.

3.4Intelligent charger technical configuration parameters

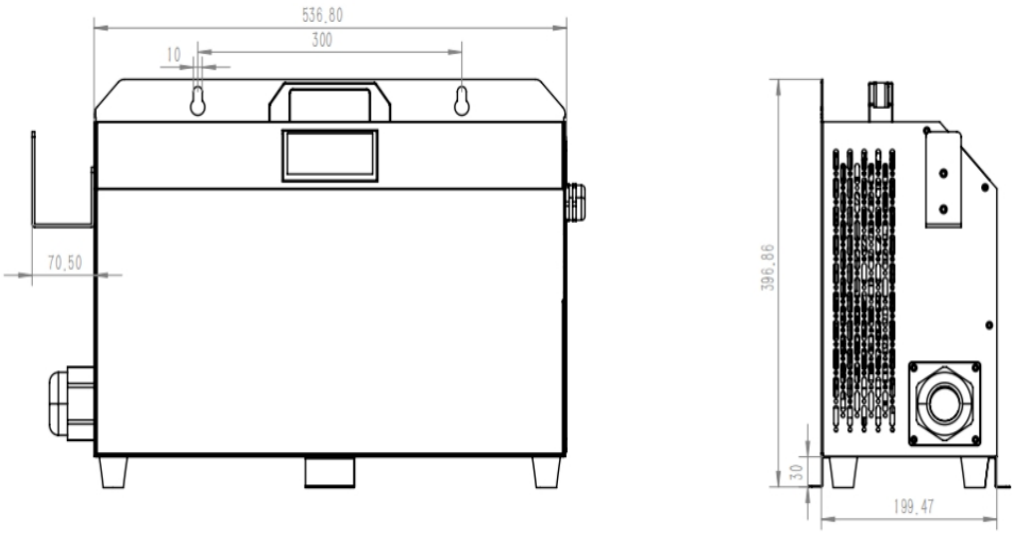
Model	80V200A	80V130A
Rated input voltage	AC380V±20% (Three-phase four-wre system)	
Input grid frequency	50/60Hz	
Power factor	≥0.99	
Input power	≤22.2KW	≤12.2KW
Efficiency	≥ 0.93	
Rated output voltage	80V	
Rated output current	200A	130A
Rated output power	20KW	11KW
DC output range	30~ 110Vdc	
Output connector	GB250A orRema320A	
Cooling method	Forced air cooling	
Working environment temperature	-20℃ to 50℃fullpower output; 50℃ to 70℃derated output	
Relative humidity	≤95%	
Protection level	IP21	IP31
Noise	<65dB	
Altitude	<2000m	

3.5 The structure and size of the charger

3.5.1 HCC80200



3.5.2 HCC80130



Chapter 4 Installation and Commissioning

4.1 Installation Precautions

The operating voltage and current of some components in the power supply system are very large. In order to ensure personal safety, the following provisions should be observed at all times:

1. Only personnel who have received power system training and have full knowledge of power systems can install this power system. The safety regulations listed above and local safety regulations should be observed at all times during installation.

2.The system voltage may reach 323Vdc or 530Vac, which is a dangerous voltage. Insulation protection is required during installation and operation to ensure personal safety.

3.If you want to operate in the power system cabinet, ensure that the power system is not powered on. Disconnect the mains input and battery power from the power system.

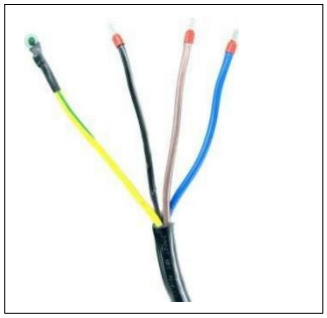
4. Place the power supply device in a locked room, lock the cabinet, and the person in charge of the power supply device will keep the key.

4.2'Installation precautions

Model	Maximum input current	Cross-sectional area of input line (mm)	Air circuit breaker model
80V200A	31A	3*6+1*6mm²	63A-3P
80V130A	18A	3*4+1*4mm²	32A-3P

4.3Enter the wiring definition ：

pencil

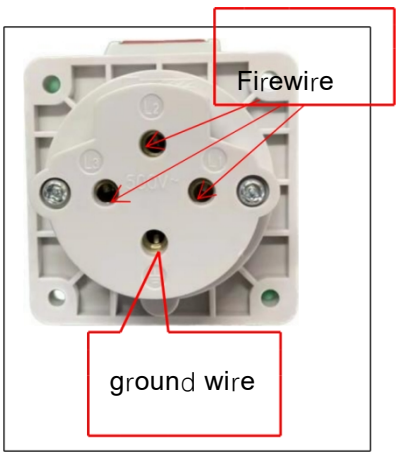


Wiring definition: The input cable is divided into three-phase 4-wire specifications, and the three-phase 4-wire interfaces are L1, L2, L3, and PE (grounded).

Explanation: L1,L2, and L3 are the three live wires, and PE is the ground wire.

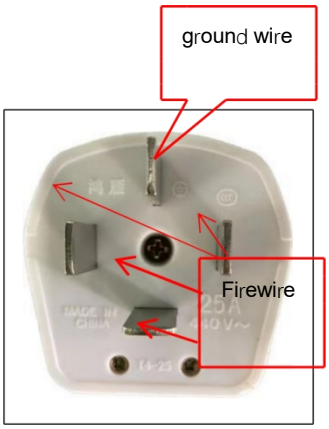
L1 shows the black line, L2 is the brown line, and L3 is the blue line. The PE is the yellow and green line, and the grounding line must be grounded.

Yeeda charging head definition



Wiring definition: Yida interface is three-phase 4 lines, R1 S2 T3 = L1 L2 L3 = A B C = UVW (four kinds of identification are national standard identification) connected to three fire lines, PE or ground wire identification grounding wire.

Installation method: the screw hole position is on the side, plug the wire into the hole, and tighten the screw from the side. 16A can connect 1-4 square lines, 32A can connect 2.5-6 square lines, 63A can connect 4-10 square lines.



Wiring definition: the interface is three-phase 4 lines, the plug front line order is: left one phase, lower two phases, right three phases, the top is the ground wire.

Warning: the PE protection land must be connected, otherwise it may threaten the safety of people and equipment. The company is not responsible for the accident caused by the user's wrong connection to the communication input line. Thank you for your cooperation.

4.4output wiring definition :

The output line is generally divided into national standard charging guns and REMA connectors, and the specific details are subject to the actual product.

National standard charging gun



Wiring definition: The charging gun is designed according to GB/T 20234.3-2015, and the wiring is defined as: DC power line: DC+, DC-, equipment ground line: PE, charging communication line: S+, S-, charging connection confirmation line: CC1, CC2, low voltage auxiliary power line: A+, A-.

REMA plug



Wiring definition: match the corresponding definition according to the customer's corresponding battery definition

4.5Installation and commissioning

Some components inside the charger have high working voltage and current.

To ensure personal safety, the following regulations should be followed:

★Only the trained and qualified personnel can install this charger.

★Insulation protection shall be provided during installation and operation.

★Ground measures for the charger should be taken.

★Fan air intake and fan air outlet should be at least 10 cm apart

★The charger should avoid the rain

★Do not bring flammable, explosive or combustible materials close to the recharger.

★Use it with caution in thunderstorm weather.

chapter5 check the maintenance instructions regularly

The input and output power wiring of the charger can only be operated by qualified professionals. Connect after the input socket is unplugged or the circuit breaker is disconnected.

1) Move the forklift near the charger and make sure the plugs and hands are dry before operation.

2) Check the charger before starting charging to ensure that the charger is wired correctly without any fault or abnormal phenomenon.

3) Confirm whether the voltage and capacity of the battery to be charged match with the charger.(Please refer to the charger specifications and the battery specifications to be charged).

4) Before charging, the output plug of the charger is connected to the socket of the forklift battery, and then the charger is input into the power plug to the power socket.

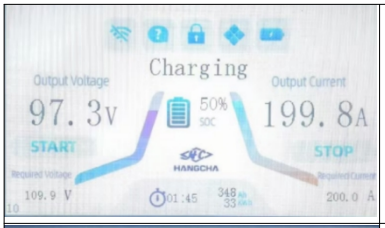
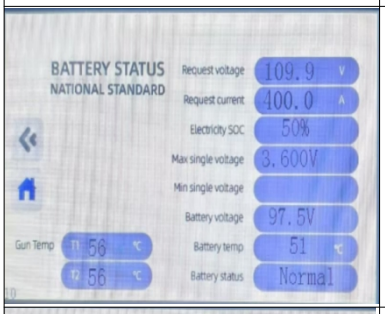
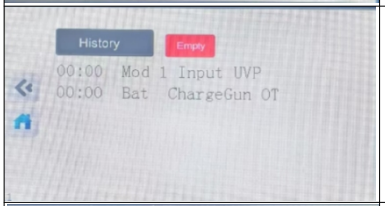
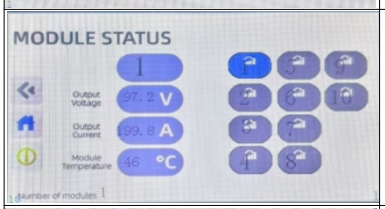
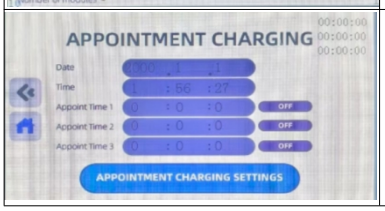
5) In the case of no fault, the charger receives the charging command (including charging voltage, charging current and startup command) issued by the BMS of the forklift battery, and automatically starts charging the battery, and the charging indicator light flashes.

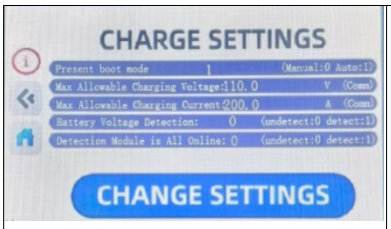
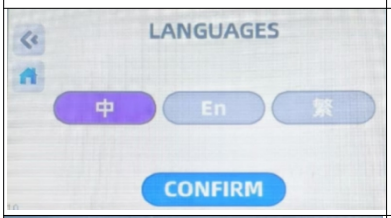
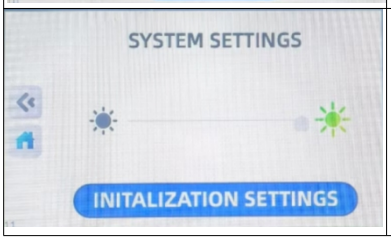
6) If an error occurs during charging, press "Pause" to stop charging until the fault is resolved.

7) After the battery is fully charged, the charger automatically stops charging, and the charger indicator light is on.

8) If you need to use a forklift when charging, press "pause" to stop charging and disconnect the plug.

9) During charging, the charging indicator blinks and the following four screens appear in sequence:

	Main interface current charging status Start, Stop, Battery SOC, Battery Temperature, Current Voltage, Current Current, Charging Reservation Countdown
	Requested current: The current setpoint value sent by the battery BMS to the charger Requested voltage: The voltage setpoint value sent by the battery BMS to the charger Battery SOC: The current state of charge value of the battery Battery temperature: The current battery temperature sent by the BMS to the charger Battery status: The current status of the battery
	History record: Records abnormal status? Note: Please clear promptly when excessive abnormalities accumulate
	Output voltage: Current output voltage of the module ? Output current: Current output current of the module ? Module temperature: Current temperature of the module
	National Standard Javelin Scheduled Charging Page: Set the scheduled charging time as countdown scheduling. For example: If charging is required after two hours, directly set it to 2.0. Note: The time unit is H. If minutes need to be set, they must be converted according to H.

	Parameter settings: For professional personnel operation only; users are for reference only.
	Language settings: Simplified Chinese, English, or Cantonese can be selected
	System Settings: Adjust Screen Brightness

10) After the charging is completed, the indicator light remains on and the display screen shows that the charging is complete. First, disconnect the AC circuit breaker of the charger, unplug the AC input plug, and finally unplug the battery connection plug.

Note: Non-standard operation will cause damage to the battery or charger, or other unexpected risks.

Chapter 6 Check the maintenance instructions regularly

★ Whether the input and output wires are damaged (test before charging);

★ Whether the output plug of the charger is damaged (test before charging);

★ If the above situation occurs, please replace the relevant accessories in time (test before charging);

★ Clean up the dust and other foreign bodies in the charger regularly (check once a month);

★ Confirm that the charger and the charger input circuit breaker are working normally. If any bad situation is found, please contact the sales store or the factory in time;

★Cut off the input power supply before the bad cause is clear, and then unplug the charging input AC plug to avoid electric shock.

Chapter 7Coding and information

touchscreen	State description	touchscreen	State description
Phase order anomaly	There is a three-phase electrical phase absence	Connector overheating	Output connector overtemperature
Mains overvoltage	Module input voltage is too high	BMS component overheating	BMS components (OTF, OTD, OTC, UTD, etc.) overtemperature
Undervoltage of mains power supply	Module input voltage too low	Connector failure	Battery and module charging connector failure
Output short circuit	Module short circuit or module output terminal short circuit	Battery temperature is too high	Battery pack temperature too high
Fan failure	The input voltage of the charger is higher or lower than the specified input voltage range.	Highvoltager elayfailure	Abnormal high-voltage battery relay (switch)
Voltage too high	Module output voltage too high	Checkpoint2 malfunction	cc2 (abnormal connection of battery t est gun port) connection abnormality
Current too high	Module output current too high	Excessive current	Battery current is too high
Reversepolarityprot ection forbatteries	The output terminals of the module are reversely connected to the positive and negative terminals of the battery.	Abnormal voltage	Abnormal battery voltage
Temperature anomaly	Module temperature is too high or too low	Other malfunctions	The state where the BMS system or equipment cannot perform its specified
Module communication exception	Abnormal communication between the module and the control board	Battery overvoltage	Battery voltage too high
Output undervoltage	The module output terminal is not connected to a battery or the battery voltage does not meet specifications.	Battery overcurrent	Battery current is too high
Insulation failure	Insulation abnormality	Battery overtemperature	Battery temperature too high

Chapter 8 Common Faults and Troubleshooting Methods

number	Fault phenomenon	Failure cause	resolvent
1	Fan fault	The fan is stuck, damaged, or the plug-in is loose.	Check the fan net cover for deformation or damage, whether the fan connector is loose, and internal accumulation of amounts of dust or foreign matter.
2	Lack of phase	Enter one phase of three phases.	Check whether the AC input voltage is normal and transmission. Check whether the entry plug is loose.
3	Input undervoltage	Input voltage is below 310V.	Check whether the AC input voltage is normal and the input plug is loose.
4	communication failure	Abnormal CAN signal line or BMS data not sent.	Check that the communication cable between the BMS and the charger is properly connected.

Chapter 9: Warranty Matters

9.1 The product is guaranteed for one year (the company does not bear the logistics, transportation, customs duties, and other costs caused by maintenance). If the warranty period expires, the user shall negotiate with the manufacturer to take a paid way for replacement or
maintenance.

9.2In the process of using our products, if there is any problem, please check according to this manual first. If still unable to resolve, please contact our company.

9.3In the process of using the company's products, if the following situations occur, the company does not assume the warranty obligation.

★ Product damage caused by use in violation of product manuals or specifications.

★ Damage to the products caused by unauthorized modification or self-repair.

★Damage to the product due to an abnormal external force.

★Damage to products due to natural disasters or other force majeure.

9.4For the quality problems caused by the scope of the warranty, the company's responsibility is limited to the product itself, and does not assume joint and extended liability.

9.5Matters needing attention when abandoned

The waste disposal of the charger is industrial waste, please hand it over to certified professionals or companies for disposal, the company is not responsible for recycling.

chapter 10 packagingshipping and storage

★ The charger is packed with carton and anti-collision material.

★Products during transportation should not be violent vibration, impact, inverted, etc.

★The charger storage should avoid high temperature and humidity, dust, metal powder and other environments, to avoid the coexistence of acidic, alkaline or other corrosive gases, stored in a dry, well-ventilated indoor environment.

★There should be pallets and other moisture proof measures on the storage floor.

HANGCHA

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